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Large Market Asymptotics for Differentiated Product Demand Estimators with Economic Models of Supply

Contents

1	Big picture	3
2	Data and measurement (model objects and estimation setup)	3
2.1	Why large-market asymptotics are the right laboratory	4
2.2	Demand side primitives and utility	4
2.3	Supply side and Bertrand-Nash pricing	4
2.4	Share inversion and the structural demand equation	5
2.5	Product-characteristic instruments	5
2.6	Asymptotic objects that matter	5
2.7	Different ways a market can become large	6
3	Models and asymptotic specifications (all equations + interpretation)	6
3.1	High-level inconsistency theorem	6
3.2	Why the simple logit benchmark already gives the intuition	7
3.3	Random-coefficients logit with single-product firms	7
3.4	Nested logit with many nests	7
3.5	Logit with many-product firms	8
3.6	Many large markets	8
3.7	Why cost instruments are structurally different	9
3.8	Recommendations for empirical practice	9
4	Identification and instruments (what makes the asymptotic argument credible)	9
4.1	What identifies price in BLP-type demand	9
4.2	Why markups are the whole first stage	10
4.3	When product-characteristic instruments fail	10
4.4	What types of variation restore instrument power	10
4.5	Why cost shifters are not implicated	10
4.6	What the paper can and cannot distinguish	11
4.7	Relation to the literature	11
5	Tables and Monte Carlo evidence	11

5.1	Monte Carlo design	11
5.2	Table I: BLP instruments with different numbers of markets and products	12
5.3	Table II: BLP instruments, ownership structure, and demand parameters	12
5.4	Table III: cost instruments as a benchmark	13
5.5	What the Monte Carlo adds to the theory	13
6	Short summary	13
7	Conclusion	14
7.1	Core conclusion (what the paper establishes)	14
7.2	Economic intuition	14
7.3	Takeaway	14

1 Big picture

- **Question.** When empirical IO researchers estimate differentiated-products demand with a small number of large markets, do standard BLP-style product-characteristic instruments still identify price sensitivity as the number of products per market gets large?
- **Setting.** The paper is theoretical. It studies discrete-choice demand estimators under Bertrand-Nash pricing and asks what happens to IV and GMM estimators when asymptotics are taken in the number of products per market rather than mainly in the number of markets.
- **Core challenge.** BLP instruments work only through equilibrium markups. If markups converge too quickly to a constant as markets get large, characteristics of rival products stop moving prices in a useful way and the first stage becomes asymptotically trivial.
- **Main conceptual contribution.** The paper connects asymptotic identification directly to the supply side. It shows that whether product-characteristic instruments are strong or weak is not just a reduced-form question about correlations in the data; it is a structural question about how markups behave in large markets.
- **Main high-level theorem.** If markups converge to a constant faster than $1/\sqrt{J}$ as the number of products J grows, then BLP instruments are asymptotically equivalent to instruments with zero correlation with prices. Estimates are inconsistent and standard confidence intervals under-cover.
- **Main negative result.** In the simple logit model and in the random-coefficients logit model without a random coefficient on price, with many products and effectively many small firms, BLP instruments fail in large markets because markups collapse toward a constant.
- **Main positive results.** BLP instruments can still work in some large-market environments: when the number of firms is fixed and products per firm grow, when the dimension of substitution patterns grows through many nests, and when the number of markets grows fast enough relative to products per market.
- **Main practical message.** There is no universal verdict on BLP instruments. Their usefulness depends on how the market becomes large and what source of variation they exploit: within-firm competition, across-firm asymmetry, or across-market size differences.
- **Main empirical recommendation.** Researchers should supplement conventional estimation with identification diagnostics, weak-IV robust confidence intervals, and Monte Carlo exercises calibrated to the intended demand-supply specification.
- **Economic takeaway.** In large differentiated-product markets, the supply side can erase the price variation that BLP instruments need. Good demand estimation therefore requires matching the instrument strategy to a structurally plausible model of markups.

2 Data and measurement (model objects and estimation setup)

2.1 Why large-market asymptotics are the right laboratory

- Many classic differentiated-products applications use only a small number of markets, each with many products.
- Berry, Levinsohn, and Pakes’s automobile application is the canonical example: about 20 markets with roughly 100 products each.
- In such settings, it is natural to ask whether adding products improves estimation or instead destroys identification by making markups too flat.

Interpretation. The paper is motivated by the empirically important case in which “sample size” comes from more products per market, not necessarily from many independent markets.

2.2 Demand side primitives and utility

The researcher observes one market with J inside goods and an outside good. Product j has price p_j , observed characteristics $x_j \in \mathbb{R}^K$, and an unobserved demand shock ξ_j . Consumer i ’s utility from good j is

$$u_{ij} = x_j' \beta - \alpha p_j + \xi_j + g(\epsilon_{ij}, \zeta_i, x_j, p_j), \quad (1)$$

where ϵ_{ij} and ζ_i are idiosyncratic demand shocks and heterogeneity terms, and σ indexes the distribution of ζ_i .

Define mean utility by

$$\delta_j \equiv x_j' \beta - \alpha p_j + \xi_j. \quad (2)$$

Each consumer buys at most one product, and market shares are population choice probabilities,

$$s_j(x, \xi, p) = E_{\epsilon, \zeta} [\mathbf{1}(u_{ij} > u_{ik} \text{ for all } k \neq j)]. \quad (3)$$

Interpretation. The whole identification problem lives in the price coefficient α and, in richer models, the heterogeneity parameter σ . Prices are endogenous because they are equilibrium outcomes and react to demand shocks through markups.

2.3 Supply side and Bertrand-Nash pricing

The paper focuses mainly on static Bertrand competition. Firm f produces the set of products F_f . Rearranging the first-order conditions for profit maximization gives

$$\sum_{k \in F_f} (p_k - MC_k) \frac{\partial s_k(x, p, \xi)}{\partial p_j} + s_j(x, p, \xi) = 0, \quad (4)$$

for every product $j \in F_f$. For single-product firms this becomes

$$p_j = MC_j - \frac{s_j(x, p, \xi)}{\frac{d}{dp_j} s_j(x, p, \xi)}. \quad (5)$$

Interpretation. BLP instruments move prices only through these equilibrium markup terms. If the supply side forces the markup to become almost constant, rival characteristics stop being useful price shifters.

2.4 Share inversion and the structural demand equation

In the models studied, shares are invertible in mean utility. Let $\delta(s, x, p, \sigma)$ denote the inverse share mapping. Then estimation is based on

$$\delta_j(s, x, p, \sigma) = x'_j \beta - \alpha p_j + \xi_j. \quad (6)$$

This is the usual BLP estimating equation.

Interpretation. Once shares are inverted, demand estimation becomes an IV problem in which price is endogenous and product characteristics plus excluded instruments must identify α and, when applicable, σ .

2.5 Product-characteristic instruments

The paper studies instruments built from characteristics of other products. Let $h_j(x_{-j})$ denote excluded functions of other products' characteristics. Then the instrument vector is

$$z_j = (x_j, h_j(x_{-j}))'. \quad (7)$$

These instruments are valid under the exclusion restriction

$$E(\xi \mid x) = 0. \quad (8)$$

The GMM criterion is

$$\left[\frac{1}{J} \sum_{j=1}^J (\delta_j(s, x, p, \sigma) - x'_j \beta + \alpha p_j) z_j \right]' W_J \left[\frac{1}{J} \sum_{j=1}^J (\delta_j(s, x, p, \sigma) - x'_j \beta + \alpha p_j) z_j \right]. \quad (9)$$

Interpretation. The question is not whether these instruments satisfy exclusion in finite samples. It is whether they still have power once the market gets large and markups flatten.

2.6 Asymptotic objects that matter

The central asymptotic object is the equilibrium markup,

$$m_j \equiv p_j - MC_j. \quad (10)$$

The paper studies whether the dependence of m_j on other products' characteristics survives as J grows.

A key benchmark comes from the simple logit model with single-product firms. There,

$$p_j = MC_j + \frac{1}{\alpha(1 - s_j)}. \quad (11)$$

If shares $s_j \rightarrow 0$, then the markup converges to $1/\alpha$.

Interpretation. Once all markups look like the same constant, rival characteristics no longer create meaningful price dispersion through competition. That is the paper’s core threat to identification.

2.7 Different ways a market can become large

The paper emphasizes that “large market” is not one thing. It considers several distinct growth paths:

- one or a few markets with many products and many small firms;
- a market with many nests, so the effective dimension of substitution patterns grows;
- a market with a fixed number of firms but many products per firm;
- many large markets, where both the number of markets and products per market increase.

Interpretation. Whether BLP instruments fail or survive depends on which of these sequences describes the data-generating process.

3 Models and asymptotic specifications (all equations + interpretation)

3.1 High-level inconsistency theorem

The paper’s main general result is Theorem 1. Suppose σ is fixed at its true value and the IV estimator uses $z_j = (x_j, h_j(x_{-j}))$. If

$$\sqrt{J} \max_{1 \leq j \leq J} |p_j - MC_j - b^*| \rightarrow_p 0 \quad (12)$$

for some constant b^* , and a standard central-limit condition holds for the sample moments, then the estimator is asymptotically equivalent to the estimator obtained from artificial prices

$$p_j^* = MC_j + b^*. \quad (13)$$

Since p_j^* differs from marginal cost only by a constant, the excluded instruments have no identifying power in this limit.

Interpretation. The theorem says: if markups converge to a constant too fast, BLP instruments become weak in the strongest possible sense. Estimation behaves as if prices had no useful instrument-driven variation beyond cost plus a constant.

3.2 Why the simple logit benchmark already gives the intuition

In the simple logit model,

$$\log s_j - \log s_0 = x'_j \beta - \alpha p_j + \xi_j, \quad (14)$$

and the markup is

$$p_j - MC_j = \frac{1}{\alpha(1 - s_j)}. \quad (15)$$

As the number of products grows and each product's share shrinks, $s_j \rightarrow 0$ and so

$$p_j - MC_j \rightarrow \frac{1}{\alpha}. \quad (16)$$

Moreover, the convergence is fast enough – of order $1/J$ under proportional shares – to trigger the inconsistency condition in Theorem 1.

Interpretation. The simple logit case shows exactly why “more products” can reduce, not increase, identification: extra competition pushes all markups toward the same limit.

3.3 Random-coefficients logit with single-product firms

The random-coefficients logit model generalizes simple logit by allowing random taste heterogeneity in observed characteristics:

$$u_{ij} = x'_j \beta - \alpha p_j + \xi_j + \sum_k x_{jk} \zeta_{ik} + \epsilon_{ij}. \quad (17)$$

The paper restricts attention to the version without a random coefficient on price. Theorem 2 shows that when products are added mainly by adding more single-product firms, markups still approach a constant quickly enough that BLP instruments fail.

Formal result. If (x_j, ξ_j, MC_j) is bounded and i.i.d. and the tails of the random coefficients are thin enough (bounded by a Gaussian tail), then condition (i) of Theorem 1 holds with

$$b^* = 1/\alpha. \quad (18)$$

Therefore, with σ known, the BLP estimator of (α, β) is inconsistent.

Interpretation. Adding smooth random heterogeneity in tastes does not rescue identification if the market becomes large by adding many small firms. The supply side still forces markups to look asymptotically constant.

3.4 Nested logit with many nests

The paper then asks how the negative result can be reversed. One route is to let the effective dimension of substitution patterns grow with the number of products while keeping the number of parameters manageable. The paper uses nested logit as the main illustration.

Main idea. When products are partitioned into more and more nests, a product's markup can

continue to depend on the composition of nearby substitutes rather than collapsing to the same constant across all goods.

Interpretation. The paper does not make nested logit the main theorem in the published text, but the message is important: BLP instruments can retain power if adding products also adds meaningful structure to demand substitution.

3.5 Logit with many-product firms

Now suppose the number of firms F is fixed while the number of products per firm increases. In the logit model, markups become constant *within* firms, not necessarily across firms. Let b_f denote the common markup for firm f . Then the equilibrium system can be written as

$$b_f = \frac{\sum_{h=0}^F e^{-\alpha b_h} \hat{\pi}_h \bar{r}_h}{\alpha \sum_{h \neq f} e^{-\alpha b_h} \hat{\pi}_h \bar{r}_h}, \quad (19)$$

where $\hat{\pi}_f$ is firm f 's product share and $\bar{r}_f$ is the average attractiveness of its products.

Theorem 3 then shows that BLP instruments can identify demand if they exploit *across-firm* variation in these objects. The estimators are asymptotically normal at rate $\sqrt{\bar{J}}$ under standard conditions.

Interpretation. In this case, product-characteristic instruments work only insofar as they capture differences across firms. Within-firm rival characteristics stop mattering, but cross-firm asymmetry still generates identifying markup variation.

3.6 Many large markets

The paper next considers N markets, each with J_i products, and lets both the number of markets and the products per market grow. Let

$$\bar{J} = \frac{1}{N} \sum_{i=1}^N J_i \quad (20)$$

be the average market size. Theorem 4 studies estimation when instruments exploit the number and composition of products in a market.

Main result.

- If $N/\bar{J} \rightarrow \infty$, then BLP-instrument estimators are consistent and asymptotically normal, with convergence rate $\sqrt{N/\bar{J}}$.
- If $N/\bar{J} \rightarrow c < \infty$, then the estimator has a weak-IV type asymptotic distribution and is inconsistent.
- If $N/\bar{J} \rightarrow 0$, the weak-IV limit collapses to the constant-markup benchmark.

The theorem also requires nontrivial variation in market sizes across markets.

Interpretation. Even if each market is large, BLP instruments can still work when the number of

markets grows fast enough and when market-size variation itself shifts markups. If instead markets become large too fast relative to the number of markets, the same weak-instrument logic returns.

3.7 Why cost instruments are structurally different

A recurring theme of the paper is that the negative results apply to instruments that shift prices *through markups*. Cost instruments are different because they can move prices directly through marginal costs. They do not require the markup to vary with rival characteristics.

Interpretation. This is why the paper’s critique is sharp but narrow. It is not a claim that differentiated-products demand is generically unidentified in large markets. It is a claim that one common instrument class can become useless under certain supply-demand asymptotics.

3.8 Recommendations for empirical practice

Section 4 translates the asymptotic theory into practical advice:

- Report weak-instrument robust confidence intervals when identification is doubtful.
- Use Monte Carlo analysis to see whether the negative asymptotics are relevant at empirically realistic market sizes.
- As a simple diagnostic, treat σ as known and run a first-stage OLS regression of price on $(x_j, h_j(x_{-j}))$; then test whether the excluded product-characteristic instruments are jointly zero.
- If a supply model is being imposed, supplement first-stage diagnostics with supply-side specification tests, because strong correlation between instruments and prices may itself indicate model misspecification if the model predicts little markup variation.

Interpretation. The paper is not telling researchers to abandon estimation. It is telling them to check whether their instrument strategy is actually compatible with the economics of their model.

4 Identification and instruments (what makes the asymptotic argument credible)

4.1 What identifies price in BLP-type demand

The identifying idea behind BLP instruments is simple: observed characteristics of rival products change the toughness of competition, which changes markups, which changes price. Exclusion comes from $E(\xi \mid x) = 0$; relevance comes from equilibrium pricing.

Interpretation. Identification is entirely mediated by how rival characteristics enter the markup. This makes the supply side central, not auxiliary.

4.2 Why markups are the whole first stage

Product characteristics of other goods do not mechanically enter the demand equation for good j as cost shifters. They matter only because firms respond to the competitive environment by changing price. Therefore, if the pricing equilibrium compresses all markups toward the same number, relevance vanishes even if exclusion remains true.

Interpretation. The paper's most important contribution is to show that weak instruments can arise endogenously from equilibrium, not only from an arbitrary local-to-zero statistical sequence.

4.3 When product-characteristic instruments fail

The failure condition is not “large markets” by itself. It is the combination of:

- a fixed or slowly growing number of markets,
- many products per market,
- markup variation that disappears faster than sampling error,
- and no strong cost instruments or alternative identifying variation.

Interpretation. The paper's negative theorem is therefore a structural warning about a specific instrument family in a specific asymptotic environment.

4.4 What types of variation restore instrument power

The paper identifies three broad ways BLP instruments can remain informative:

- **Across-firm asymmetry:** with a fixed number of firms and many products per firm, firm-level differences sustain markup differences.
- **Richer substitution patterns:** many nests or other structured random-coefficient growth can prevent markups from collapsing to one constant.
- **Across-market variation:** if the number of markets grows quickly and market size varies sufficiently, BLP instruments can use that cross-market heterogeneity.

Interpretation. The paper does not say that researchers need cost shifters in every application. It says they need some source of variation that the equilibrium model says will still move markups in large samples.

4.5 Why cost shifters are not implicated

If instruments shift marginal cost directly, they do not rely on markup dispersion. A supply model can therefore remain identified even when product-characteristic instruments fail.

Interpretation. The critique is strongest for applications that lean heavily or exclusively on BLP-style functions of rival characteristics and have little cost-side information.

4.6 What the paper can and cannot distinguish

- It **can** show when the economics of large-market pricing imply weak or trivial relevance of BLP instruments.
- It **can** compare several growth paths for market size and show that they have sharply different identification implications.
- It **cannot** settle every demand model of practical interest. In particular, the main random-coefficients result excludes a random coefficient on price, and extending the theorems to richer random-coefficient structures is left for future work.
- It **does not** solve the full finite-sample inference problem analytically; instead, it recommends robust inference and Monte Carlo evidence.

4.7 Relation to the literature

The paper sits at the intersection of three literatures:

- differentiated-products demand estimation after BLP;
- weak-instrument asymptotics;
- oligopoly pricing in large discrete-choice markets.

Its novel step is to derive weak-IV sequences from equilibrium markups rather than simply positing them statistically.

5 Tables and Monte Carlo evidence

5.1 Monte Carlo design

The main quantitative evidence is not a data application but a Monte Carlo study built around the random-coefficients logit model from Section 3.1.

- Prices are generated from Bertrand equilibrium.
- Depending on the design, markets have 20, 60, or 100 products, or a mixture of those values.
- When products per firm vary, one-third of markets have 2, one-third have 5, and one-third have 10 products per firm.
- Product characteristics include a constant and a uniform $(0, 1)$ variable.
- There is an excluded cost shifter, also uniform, and marginal cost is linear in observables plus an unobserved component.
- The random coefficient on the nonprice characteristic is Gaussian with variance $\sigma^2 = 9$.
- The true price coefficient is $\alpha = 1$.

Interpretation. The simulation asks whether the paper’s asymptotic logic shows up at market sizes that applied researchers would actually recognize as realistic.

5.2 Table I: BLP instruments with different numbers of markets and products

Read:

- With only 1 market and 60 or 100 products per market, the BLP estimator performs very poorly. Median bias of $\hat{\alpha}$ is about -0.37 , median absolute deviation is around 0.67 to 0.72, and power against $\alpha = 0$ is only about 0.10 to 0.14.
- With 3 markets and only 20 products per market, the estimator improves but is still noisy: median bias is about -0.11 and power is roughly 0.40.
- With 20 markets and 20 products per market, performance becomes excellent: median bias is essentially zero (0.0039) and power rises to 0.988.
- With 20 markets but 100 products per market, performance deteriorates again: power falls to 0.456 and median absolute deviation rises to 0.315.

Economic message. The problem is exactly the one highlighted by the theory. BLP instruments break down when products per market are large relative to the number of markets, and they recover when the number of markets is large enough relative to market size.

TABLE I
MONTE CARLO RESULTS FOR BLP INSTRUMENTS (10 PRODUCTS PER FIRM, $\beta = (3, 6)$)

Markets	Products per Market	Median Bias of $\hat{\alpha}$	Median Absolute Deviation From α_0	Rejection Prob. at True α	Power of Test of $\alpha = 0$
1	60	-0.3698	0.6691	0.0783	0.1004
1	100	-0.3648	0.7177	0.1211	0.1381
3	20	-0.1053	0.3358	0.0390	0.3980
3	Varied	-0.0494	0.2966	0.1523	0.4649
3	60	-0.2186	0.5827	0.0200	0.1040
3	100	-0.2525	0.6383	0.1351	0.1762
20	20	0.0039	0.1140	0.0390	0.9880
20	Varied	-0.0014	0.1001	0.0400	0.9960
20	60	0.0130	0.2345	0.0230	0.7710
20	100	-0.0379	0.3154	0.0200	0.4560

TABLE II
MONTE CARLO RESULTS FOR BLP INSTRUMENTS (20 MARKETS, 100 PRODUCTS PER MARKET)

β	Products per Firm	Median Bias	Median Absolute Deviation From α_0	Rejection Prob. at True α	Power of Test of $\alpha = 0$
(3, 1)	10	-0.2928	0.8090	0.1012	0.1273
(3, 1)	2	-0.3573	0.7653	0.0991	0.1241
(3, 1)	Varied	-0.0054	0.1895	0.0380	0.8440
(3, 6)	2	-0.2191	0.6897	0.1061	0.1632
(3, 6)	Varied	-0.0190	0.1659	0.0410	0.9450

5.3 Table II: BLP instruments, ownership structure, and demand parameters

Read:

- Table II fixes 20 markets and 100 products per market and varies the ownership structure and coefficient values.
- When products per firm are fixed at 10, median bias is about -0.29 and power is only 0.13 for one calibration.
- When products per firm vary across markets, the estimator improves dramatically: for $\beta = (3, 1)$, median bias becomes about -0.005 and power rises to 0.844; for $\beta = (3, 6)$, median bias

TABLE III
MONTE CARLO RESULTS FOR COST INSTRUMENTS (10 PRODUCTS PER FIRM, $\beta = (3, 6)$)

Markets	Products per Market	Median Bias of $\hat{a}$	Median Absolute Deviation From α_0	Rejection Prob. at True α	Power of Test of $\alpha = 0$
1	60	-0.0247	0.1749	0.1130	0.6710
1	100	-0.0196	0.1358	0.0762	0.7623
3	20	-0.0262	0.1837	0.1002	0.6092
3	Varied	-0.0122	0.1000	0.0852	0.7916
3	60	-0.0102	0.1007	0.0661	0.7768
3	100	-0.0054	0.0767	0.0662	0.8175
20	20	0.0065	0.0663	0.0220	0.7840
20	Varied	-0.0008	0.0385	0.0522	0.8554
20	60	-0.0023	0.0365	0.0641	0.8707
20	100	-0.0027	0.0298	0.0481	0.8826

becomes about -0.019 and power rises to 0.945 .

Economic message. The same overall market size can generate very different identification quality depending on whether markup variation survives through firm structure. This is the simulation counterpart to Theorem 3.

5.4 Table III: cost instruments as a benchmark

Read:

- Cost-instrument estimates are stable across all designs. Median bias ranges roughly from -0.025 to 0.007 .
- Median absolute deviation is much smaller than for BLP instruments and falls as the number of markets grows.
- Power against $\alpha = 0$ remains respectable even in difficult designs, ranging from about 0.61 to 0.88 .

Economic message. The Monte Carlo confirms the paper’s structural distinction: the large-market problem is about instruments that rely on markup variation, not about demand estimation per se.

5.5 What the Monte Carlo adds to the theory

The theory gives asymptotic conditions. The tables show that those asymptotics are already relevant in finite samples of practical importance, such as tens of markets and dozens or hundreds of products.

Interpretation. This is why the paper’s message is operational. A researcher does not need an astronomically large market for the weak-IV problem to matter.

6 Short summary

- BLP instruments identify price only through equilibrium markups.

- In several important large-market sequences, especially many products with many small firms, markups converge too quickly to a constant and BLP instruments become asymptotically useless.
- The paper’s negative result is strongest for simple logit and random-coefficients logit without random coefficients on price.
- Identification can survive if products are added in ways that preserve markup variation: across firms, across nests, or across sufficiently many markets.
- Cost instruments are not subject to the same problem because they shift prices directly.
- In practice, the paper recommends weak-IV robust inference, first-stage diagnostics, and Monte Carlo checks tied to the intended supply-demand model.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that large-market asymptotics can fundamentally change the identification properties of differentiated-products demand estimators. Product-characteristic instruments that look natural in finite samples may become asymptotically trivial once the supply side drives markups toward a constant.

7.2 Economic intuition

The central economic force is simple: BLP instruments do not move prices by magic. They move prices only because rival characteristics change the toughness of competition and therefore the markup. If large markets flatten the markup schedule, the first stage disappears. If some form of asymmetry survives, identification can survive with it.

7.3 Takeaway

The paper’s lesson is methodological and substantive. Methodologically, researchers must tie instrument choice to the equilibrium model of pricing, not just to exclusion stories. Substantively, how a market becomes large – through more firms, more products per firm, richer substitution, or more markets – determines whether standard BLP-style estimation remains reliable.